

LAPORAN PRAKTIKUM

BASIS DATA

MODUL 13



Disusun Oleh :

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**PROGRAM STUDI REKAYASA
FAKULTAS INFORMATIKA**

**TELKOM UNIVERSITY PURWOKERTO
2025**

1. INSERT SATU TOKO

```
> db.toko.insertOne({
  "name" : "Toko Nusantara",
  "category" : "Elektronik",
  "city" : "Semarang",
  "address" : {
    "street" : "Jl. Pemuda",
    "zipcode" : "50132"
  },
  "products" : [
    { "item" : "Mouse Gaming", "price" : 350000, "stock" : 15 },
    { "item" : "Keyboard RGB", "price" : 750000, "stock" : 8 }
  ]
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a1301dc6125a8fc9b9636b3')
}
```

2. TAMPILKAN TOKO DI JKT

```

> db.toko.find({ city: "Jakarta" })
< {
  _id: ObjectId('6a1301b86125a8fc9b9636af'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
{
  _id: ObjectId('6a1301b86125a8fc9b9636b1'),
  name: 'Toko Salim Bakery',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {

```

3. PRODUK HARGA LEBIH DARI 3JT

The screenshot displays the MongoDB Compass application. On the left sidebar, the 'Connections' panel shows two connections: 'bebas' and 'JM13'. The 'JM13' connection is expanded, revealing a tree structure with 'admin', 'config', 'local', and 'providers' databases, and a 'providers' collection. The main workspace is titled 'mongosh: JM13' and contains a query editor and a results pane.

The query editor shows the following JavaScript code:

```
>_MONGOSH
}
> db.toko.find({
  "products.price": { $gt: 3000000 }
})
< {
  _id: ObjectId('6a1301b86125a8fc9b9636af'),
  name: 'Toko Berkah',
  category: 'Elektronik',
  city: 'Jakarta',
  address: {
    street: 'Jl. Sudirman',
    zipcode: '10220'
  },
  products: [
    {
      item: 'TV',
      price: 3500000,
      stock: 10
    },
    {
      item: 'Kulkas',
      price: 2800000,
      stock: 5
    }
  ]
}
```

The results pane shows the first document returned by the query, which is a JSON object representing a store named 'Toko Berkah' located in Jakarta. It has an 'Elektronik' category and a list of products. The first product is a 'TV' with a price of 3,500,000 and a stock of 10. The second product is a 'Kulkas' (refrigerator) with a price of 2,800,000 and a stock of 5.

4. UBAH KATAGORI DAN TAMBAH PRODUK BARU

My Queries

Data Modeling

CONNECTIONS (2)

×

+

...

Search connections

▼

▶ bebas

▼ JM13

- ▶ admin
- ▶ config
- ▶ local
- ▼ providers
 - providers

>_MONGOSH

}

> db.toko.updateOne(

{ name: "Toko Berkah" },

{ \$set: { category: "Aksesoris" } }

)

< {

acknowledged: true,

insertedId: null,

matchedCount: 1,

modifiedCount: 1,

upsertedCount: 0

}

> db.toko.updateOne(

{ name: "Toko Maju" },

{

\$push: {

products: {

item: "Printer Epson",

price: 25000000,

stock: 6

< {

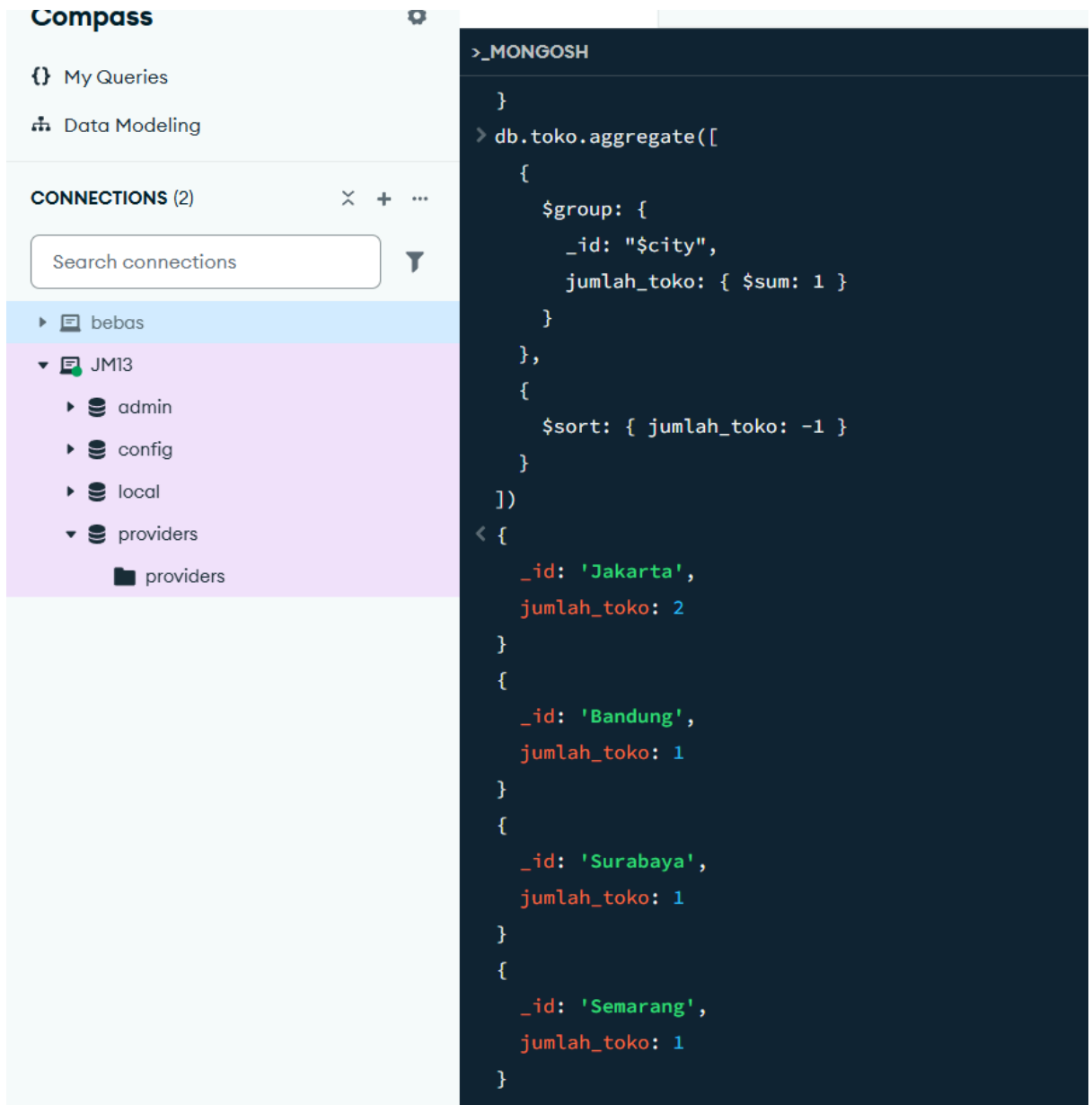
acknowledged: true,

insertedId: null,

matchedCount: 1,

modifiedCount: 1,

5. HITUNG JUMLAH TOKO PER KOTA



The image shows the MongoDB Compass interface on the left and a terminal window on the right. The Compass interface displays a connection named 'JM13' with a tree view showing 'admin', 'config', 'local', and 'providers' databases. The terminal window shows a MongoDB aggregation query being executed in the MONGOSH shell.

Compass CONNECTIONS (2)

- bebas
- JM13
 - admin
 - config
 - local
 - providers

>_MONGOSH

```
}
> db.toko.aggregate([
  {
    $group: {
      _id: "$city",
      jumlah_toko: { $sum: 1 }
    }
  },
  {
    $sort: { jumlah_toko: -1 }
  }
])
< {
  _id: 'Jakarta',
  jumlah_toko: 2
}
{
  _id: 'Bandung',
  jumlah_toko: 1
}
{
  _id: 'Surabaya',
  jumlah_toko: 1
}
{
  _id: 'Semarang',
  jumlah_toko: 1
}
}
```

6. RATA RATA HARGA PRODUK PER KATAGORI

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists two connections: 'bebas' and 'JM13'. The 'JM13' connection is expanded, showing a tree view of databases: 'admin', 'config', 'local', and 'providers'. The 'providers' database is selected. The main panel on the right shows the MongoDB shell with the following aggregate query:

```
> db.toko.aggregate([
  { $unwind: "$products" },
  {
    $group: {
      _id: "$category",
      rata_rata_harga: { $avg: "$products.price" }
    }
  },
  {
    $sort: { rata_rata_harga: -1 }
  }
])
```

The query results are displayed below the shell, showing two documents:

```
< {
  _id: 'Elektronik',
  rata_rata_harga: 5127777.777777778
}
{
  _id: 'Aksesoris',
  rata_rata_harga: 3150000
}
```

7. TAMBAH PRODUK KE TOKO MANJU

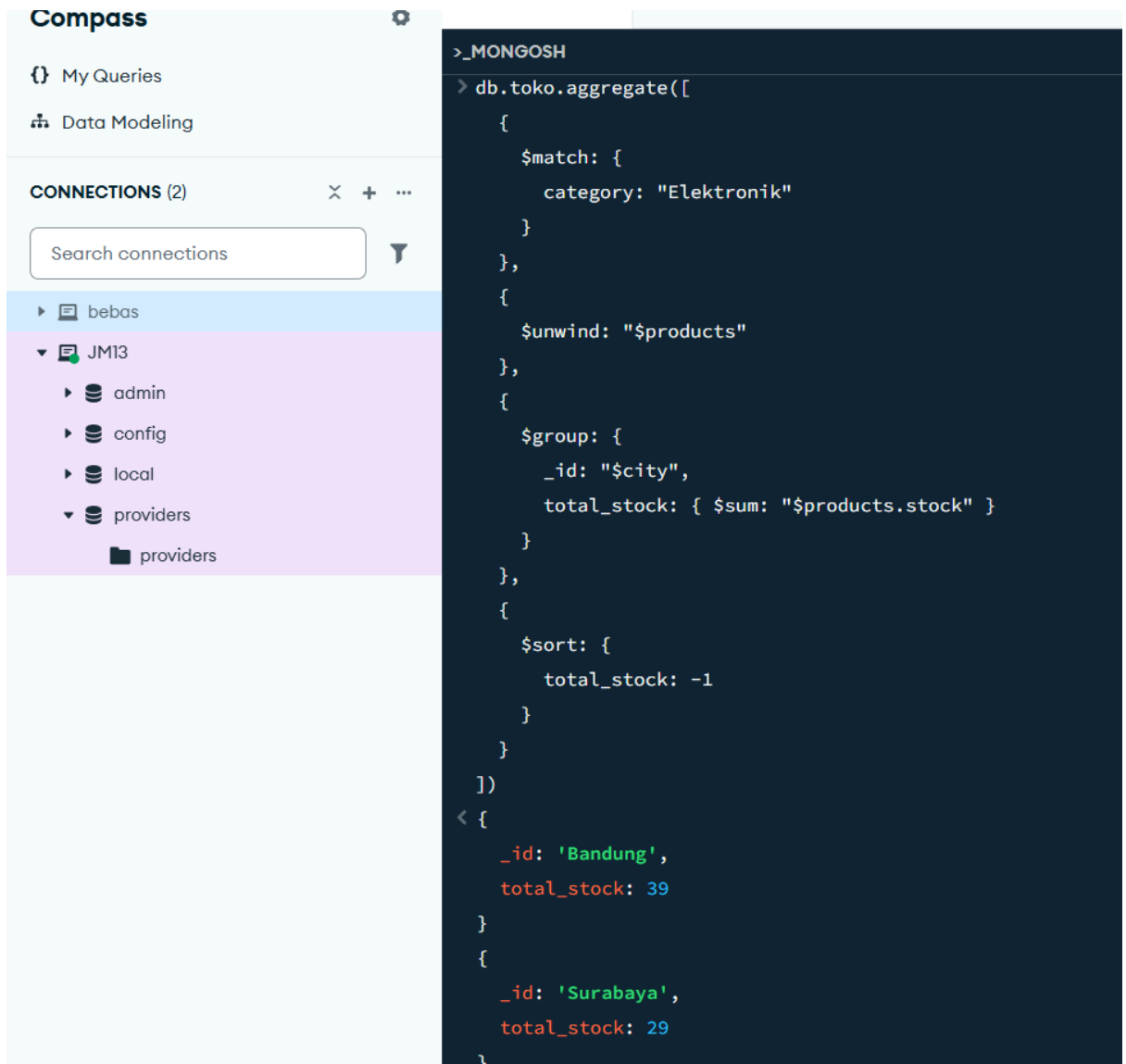
```
> db.toko.updateOne(
  { name: "Toko Maju" },
  {
    $push: {
      products: {
        item: "Speaker JBL",
        price: 1200000,
        stock: 10
      }
    }
  }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

8. HARGA PRODUK TERTINGGI PER CITY

The screenshot displays the MongoDB Compass application. On the left sidebar, under 'CONNECTIONS (2)', there is a search bar and a list of connections. The 'JM13' connection is expanded, showing a tree structure with 'admin', 'config', 'local', and 'providers' databases. The 'providers' database is currently selected. On the right, the MongoDB shell is open, showing a MongoDB aggregation pipeline. The pipeline uses the `$unwind` operator to flatten the `products` array, followed by a `$group` stage to calculate the maximum price (`$max: '$products.price'`) for each city (`city`). The output shows four documents, one for each city: Bandung (8000000), Jakarta (7700000), Surabaya (12000000), and Semarang (750000).

```
>_MONGOSH
}
> db.toko.aggregate([
  { $unwind: "$products" },
  {
    $group: {
      _id: "$city",
      harga_tertinggi: { $max: "$products.price" }
    }
  }
])
< {
  _id: 'Bandung',
  harga_tertinggi: 8000000
}
{
  _id: 'Jakarta',
  harga_tertinggi: 7700000
}
{
  _id: 'Surabaya',
  harga_tertinggi: 12000000
}
{
  _id: 'Semarang',
  harga_tertinggi: 750000
}
```

9. PIPELINE LENGKAP



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists two connections: 'bebas' and 'JMI3'. Under 'JMI3', there are sub-connections for 'admin', 'config', 'local', and 'providers'. The 'providers' connection is selected. On the right, the MongoDB Shell shows an aggregate query for the 'db.toko' collection. The query filters for 'category: "Elektronik"', unwinds the '\$products' array, groups by '_id: "\$city"' and sums the 'total_stock' from '\$products.stock', and sorts by 'total_stock' in descending order. The result shows two documents: one for 'Bandung' with a total stock of 39, and one for 'Surabaya' with a total stock of 29.

```
>_MONGOSH
> db.toko.aggregate([
  {
    $match: {
      category: "Elektronik"
    }
  },
  {
    $unwind: "$products"
  },
  {
    $group: {
      _id: "$city",
      total_stock: { $sum: "$products.stock" }
    }
  },
  {
    $sort: {
      total_stock: -1
    }
  }
])
< {
  _id: 'Bandung',
  total_stock: 39
}
{
  _id: 'Surabaya',
  total_stock: 29
}
```

10. HAPUS TOKO DENGAN KATEGORI FASION DI SBY

```
> db.toko.deleteOne({
  category: "Fashion",
  city: "Surabaya"
})
< {
  acknowledged: true,
  deletedCount: 0
}
providers> |
```